

A GraphCast-style fitting model as Plexus2 operators

prototype/graphcast — nothing promoted

1. Mechanism

Every model in `connectome-gnn` is a one-step message-passing network on a *given* connectome. Two things wanted now sit outside that: a deeper, edge-stateful processor (GraphCast’s), and datasets with no connectome at all. ZAPBench has 71,721 neurons of which about 481 are matched to an electron-microscopy reconstruction; the redox organoid is a three-dimensional field with no segmentation. In both, the coupling has to come from *space* rather than from wiring.

The arithmetic that decides what can be represented: a ZAPBench volume is acquired plane by plane, 72 planes at 12.0 ms, so one volume takes 0.851 s against a 0.914 s frame interval — 93% of the frame period. Neurons at different depths are therefore *not* sampled simultaneously, which is precisely what a message-passing operator assumes. That fact is data, not a modelling choice: the per-plane acquisition times are recorded, and the model is conditioned on the offset rather than pretending it away.

2. Equations and symbols

To be written at stage 2, when the operator exists. The two message forms this prototype selects between are

$$\text{simple} \quad \mathbf{m}_i = \sum_j W_{ij} g_\phi(v_j, a_j), \quad \dot{v}_i = f_\theta(v_i, a_i, \mathbf{m}_i, b_i s(t)) \quad (1)$$

$$\text{graphcast} \quad e_{ij} \leftarrow e_{ij} + \text{LN}(\text{MLP}_e([e_{ij} \| v_j \| v_i])), \quad v_i \leftarrow v_i + \text{LN}(\text{MLP}_v([v_i \| \sum_j e_{ij}])) \quad (2)$$

with a_i the per-neuron embedding carrying heterogeneity invisible in the state, and b_i the per-neuron gain on the external stimulus $s(t)$.

3. The Plexus2 container

To be written as the operators land. Sets: `neuron`. Fields: `stim` (playback), `mesh` (only when the encoder/decoder option is on). The four architectural choices are *config options*, not forks — `encoder_decoder`, `message`, `n_passes`, `embedding` — which is why the gate table below counts 24 combinations.

4. Gates

A gate is a number with a threshold decided *before* the run and a stated consequence if it fails. A threshold chosen after seeing the number is not a threshold, so every threshold below is a literal in `gates.py` and predates the code it gates. Three tiers, and only the third is a statement about biology; this table is 9 bookkeeping / 13 closed form / 4 measurement, so two thirds of it is verification. That is the expected shape and is stated rather than hidden.

A gate’s threshold belongs in the unit of the phenomenon, not of the mesh — a threshold in grid cells is denominated in the currency in which it is easiest to pass, and a green row of that kind

reads as an endorsement the interface has not earned. Every row therefore carries what its number is *of*, and the measurement tier is only available because the spec declares `general.units:`.

gate		threshold	measured	outcome	figure
<i>bookkeeping — does the code do what the operator says?</i>					
G1	all four options PARSE (24 combinations)	24 of 24 option combinations load	24	PASS	G1_options.png
G1b	all four options BUILD and take one step	24 of 24 option combinations run one forward step	—	SKIP	—
G2	nothing dataset-specific is hardcoded	0 of-fending literals outside config/	0	PASS	G2_scan.png
G3	scatter->gather round trip on a constant field	< 1e-6 of the field value	—	SKIP	—
G4	transfer weights are a partition of unity	sum(w) - 1 < 1e-6	—	SKIP	—
G5	simple + 1 pass + no enc/dec is arithmetically NeuralGNN	< 1e-5 of the voltage range	—	SKIP	—
G6	residual blocks start at identity: 1 vs 16 passes at init	bit-identical (max delta == 0)	—	SKIP	—
G7	units declared, and every measurement threshold is in phenomenon units	1 = declared and checked	1	PASS	G7_units.png
G8	a K=20 rollout on the toy does not diverge	state norm stays < 2x the ground-truth norm	—	SKIP	—
<i>closed form — does it reproduce the physics it was given?</i>					
G9	recover the spatial interaction kernel (toy_small)	$R^2 > 0.90$ against the GT kernel	—	SKIP	—
G10	recover per-type time constants, GT graph supplied	$R^2 > 0.95$ against the known tau	—	SKIP	—
G11	a_i recovers the types (toy_small)	ARI > 0.70 against the 8 true types	—	SKIP	—
G12	a_i recovers the types (toy_large)	ARI > 0.70 against the 65 true types	—	SKIP	—
G13	recover the per-neuron stimulus gain b_i	$R^2 > 0.90$ against the true gain	—	SKIP	—
G14	encoder/decoder is a genuine option: on vs off	delta R^2(kernel) < 0.03	—	SKIP	—
G15	graphcast vs simple message is RESOLVED, either way	delta re-reported	—	SKIP	—

What the gates license, so far. Only stage 0 has run. G1 establishes that all 24 option combinations parse, G2 that no dataset path, name or dimension appears anywhere in the prototype’s code, and G7 that the spec declares its units — so the measurement-tier rows below are *available* to be denominated in seconds and micrometres rather than in cells. Nothing about any model is licensed yet: no operator has been built, and every closed-form and measurement row is still **SKIP**. G5 is the one to watch, because it is what will establish that the **simple** option is arithmetically the existing **NeuralGNN** and therefore that everything after it is a controlled variation on something already known to work.

5. The evidence

Every gate above points to a figure, and a gate with no artifact is reported as **SKIP** rather than passed. That is not bookkeeping for its own sake: three defects in the toy were found this way and each was a number that looked plausible — a circuit sitting at a fixed point (spatial spread 9.29 against a temporal spread of 0.065), a stimulus field that was *identically zero* for three generations because **activation_pulse** reads a clock no operator was writing, and a spatial type purity of $6.1 \times$ chance that was the grid being finer than the sampling. All three are obvious in a panel and invisible in a scalar.

a option combinations					
off/simpl/p1/none	off/simpl/p1/free	off/simpl/p1/mult	off/graph/p1/none	off/graph/p1/free	off/graph/p1/mult
off/graph/p4/none	off/graph/p4/free	off/graph/p4/mult	off/graph/p16/none	off/graph/p16/free	off/graph/p16/mult
on/simpl/p1/none	on/simpl/p1/free	on/simpl/p1/mult	on/graph/p1/none	on/graph/p1/free	on/graph/p1/mult
on/graph/p4/none	on/graph/p4/free	on/graph/p4/mult	on/graph/p16/none	on/graph/p16/free	on/graph/p16/mult
24/24 parse					

Figure 1: **G1** — all four options PARSE (24 combinations). Threshold: 24 of 24 option combinations load. Measured: 24. Outcome: PASS.

G16 also has a movie: [G16_state.mp4](#).

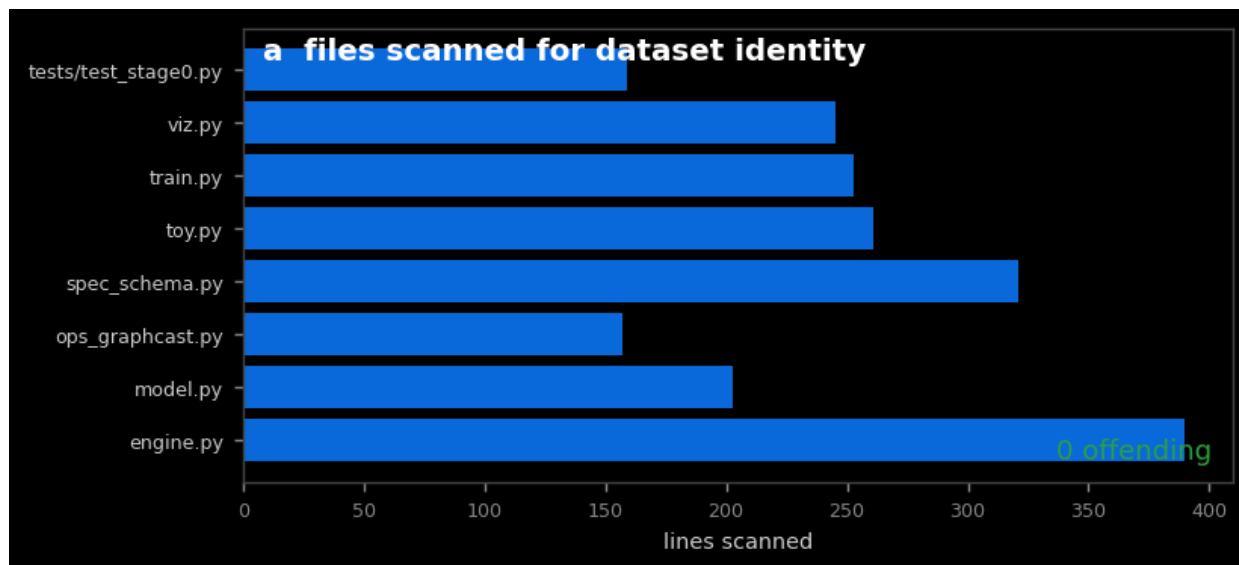


Figure 2: **G2** — nothing dataset-specific is hardcoded. Threshold: 0 offending literals outside config/. Measured: 0. Outcome: PASS.

a declared scale, and what it makes available	
length_um	100 um per length unit
time_s	0.05 s per time unit
force_nN	ratios only
-> rate	20 per s
-> velocity	2000 um/s

Figure 3: **G7** — units declared, and every measurement threshold is in phenomenon units. Threshold: 1 = declared and checked. Measured: 1. Outcome: PASS.

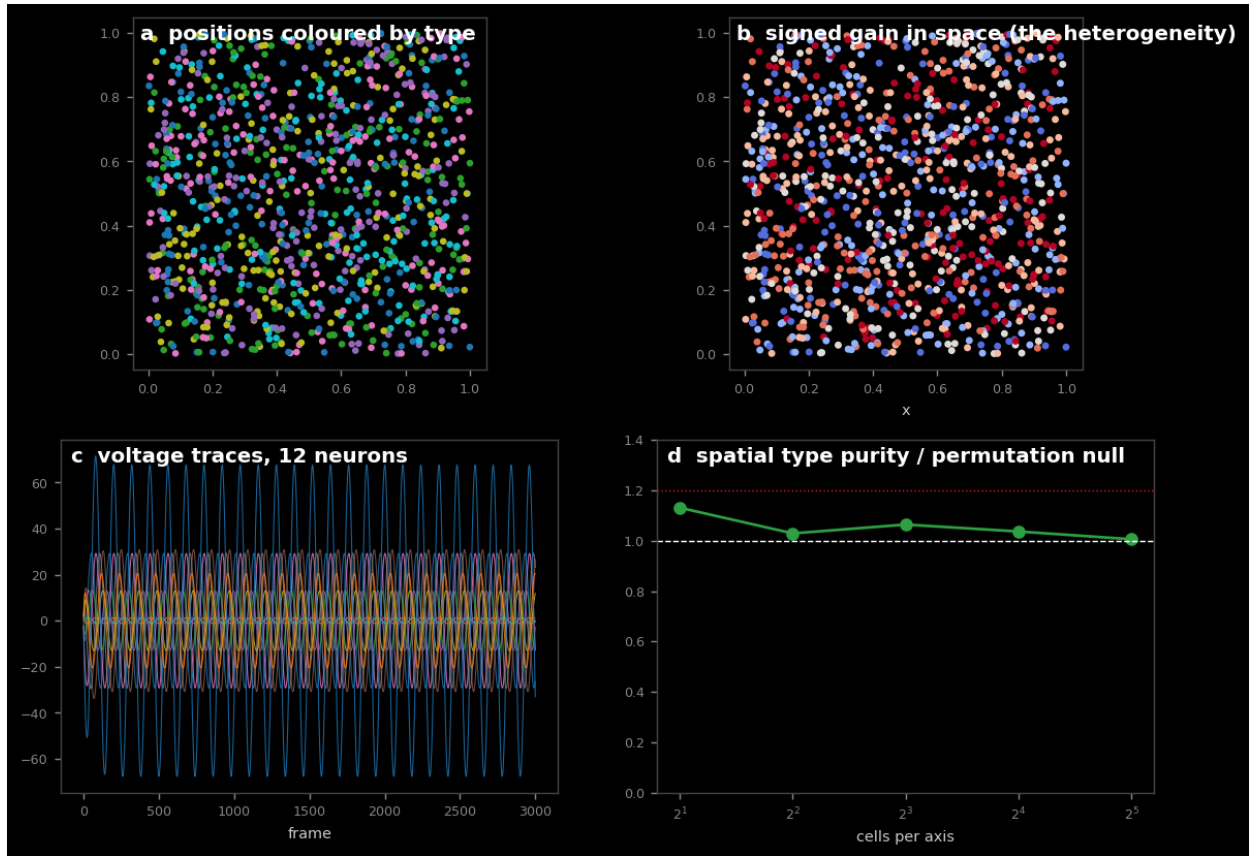


Figure 4: **G16** — types are spatially mixed by construction. Threshold: spatial-cell type purity within 20% of chance ($1/n_types$). Measured: 1.131. Outcome: PASS.

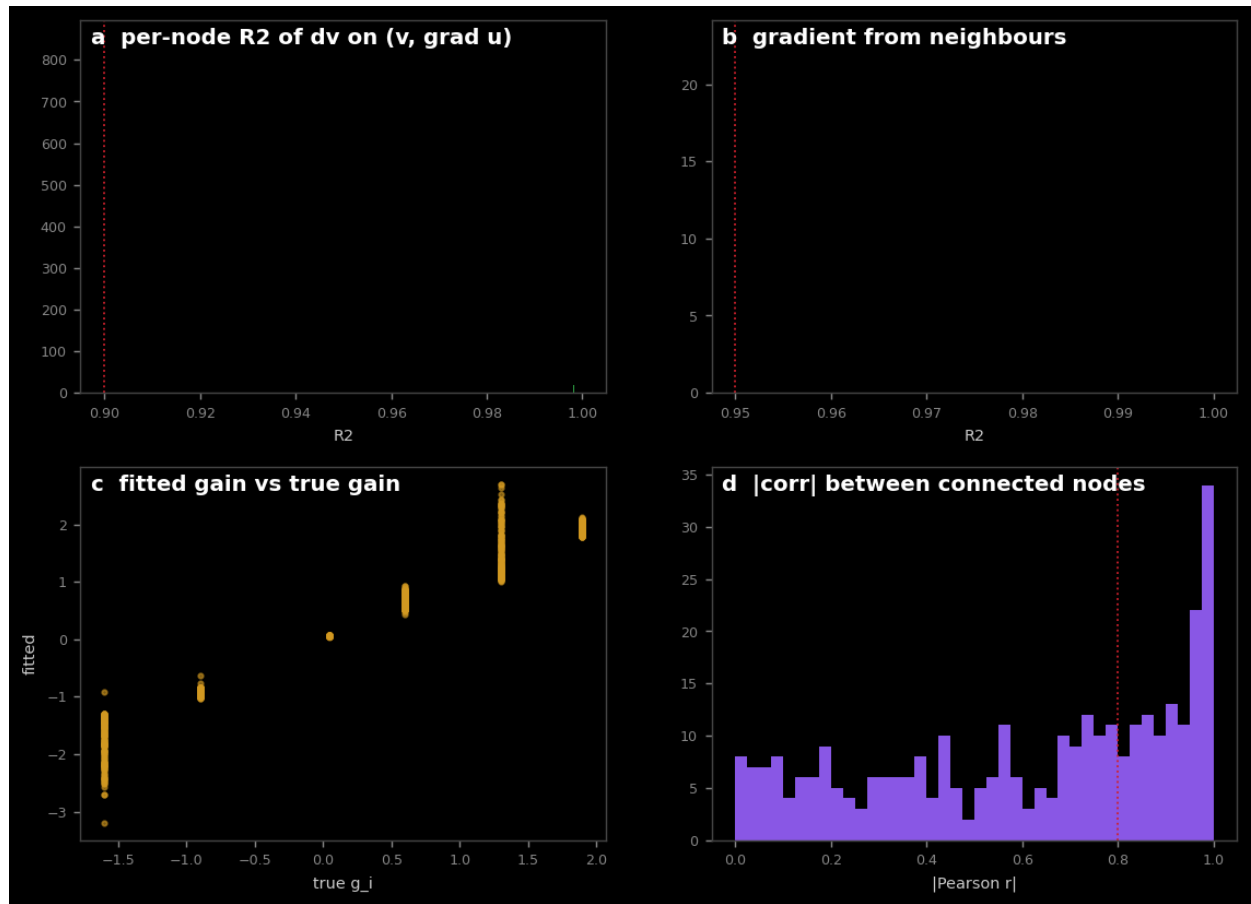


Figure 5: **G22** — the fine rule is exactly recoverable from $(v, \text{grad } u)$. Threshold: minimum per-node $R^2 > 0.90$. Measured: 0.9981. Outcome: PASS.

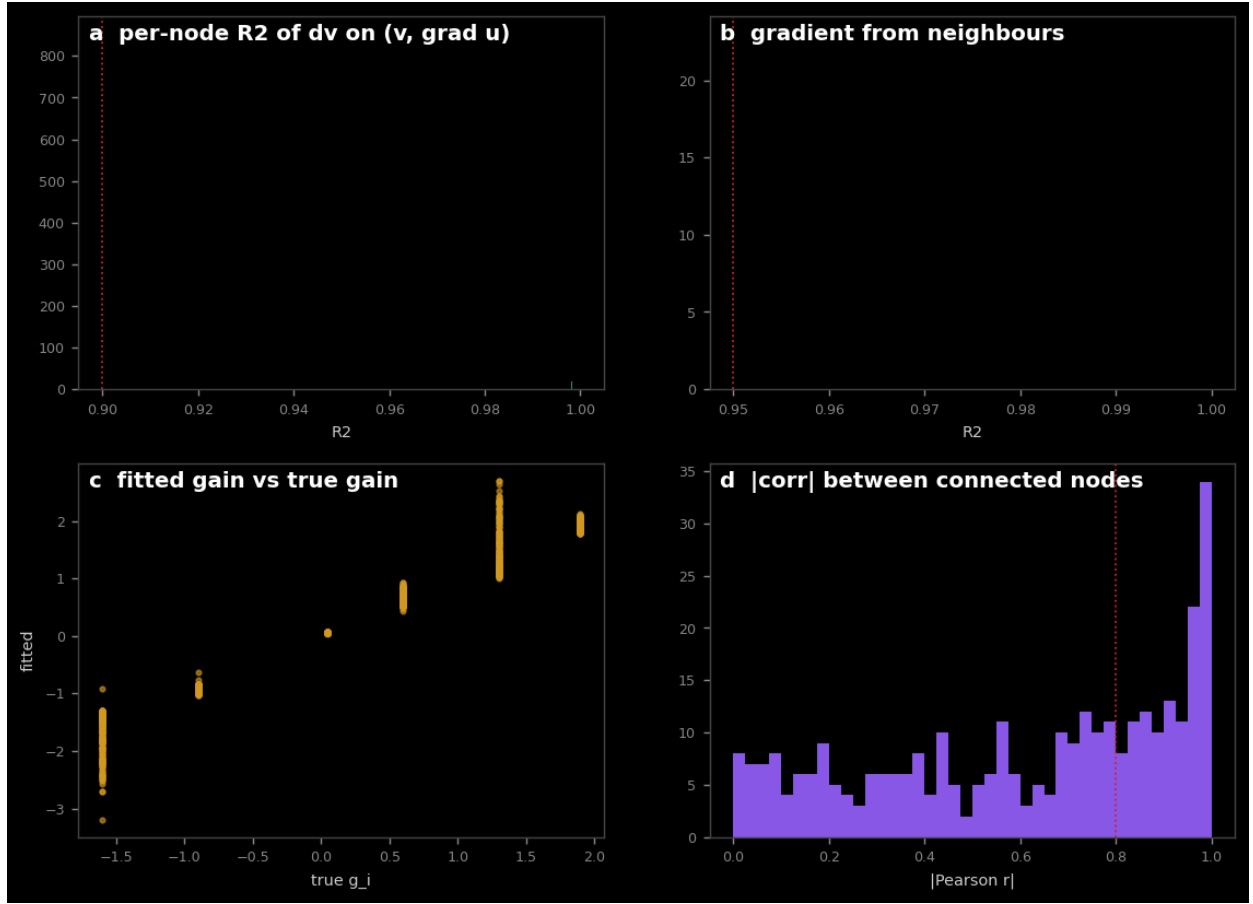


Figure 6: **G23** — the gradient is reconstructible from neighbours' states. Threshold: $R^2 > 0.95$, else the graph cannot carry the fine rule. Measured: 1. Outcome: PASS.

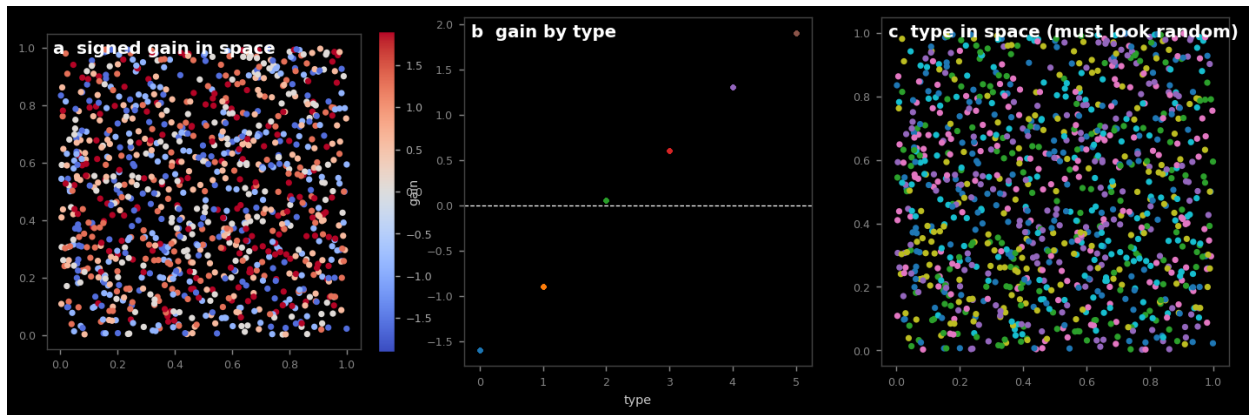


Figure 7: **G24** — the heterogeneity is linearly readable. Threshold: $\text{corr}(\text{fitted gain}, \text{true } g_i) > 0.90$. Measured: 0.9807. Outcome: PASS.

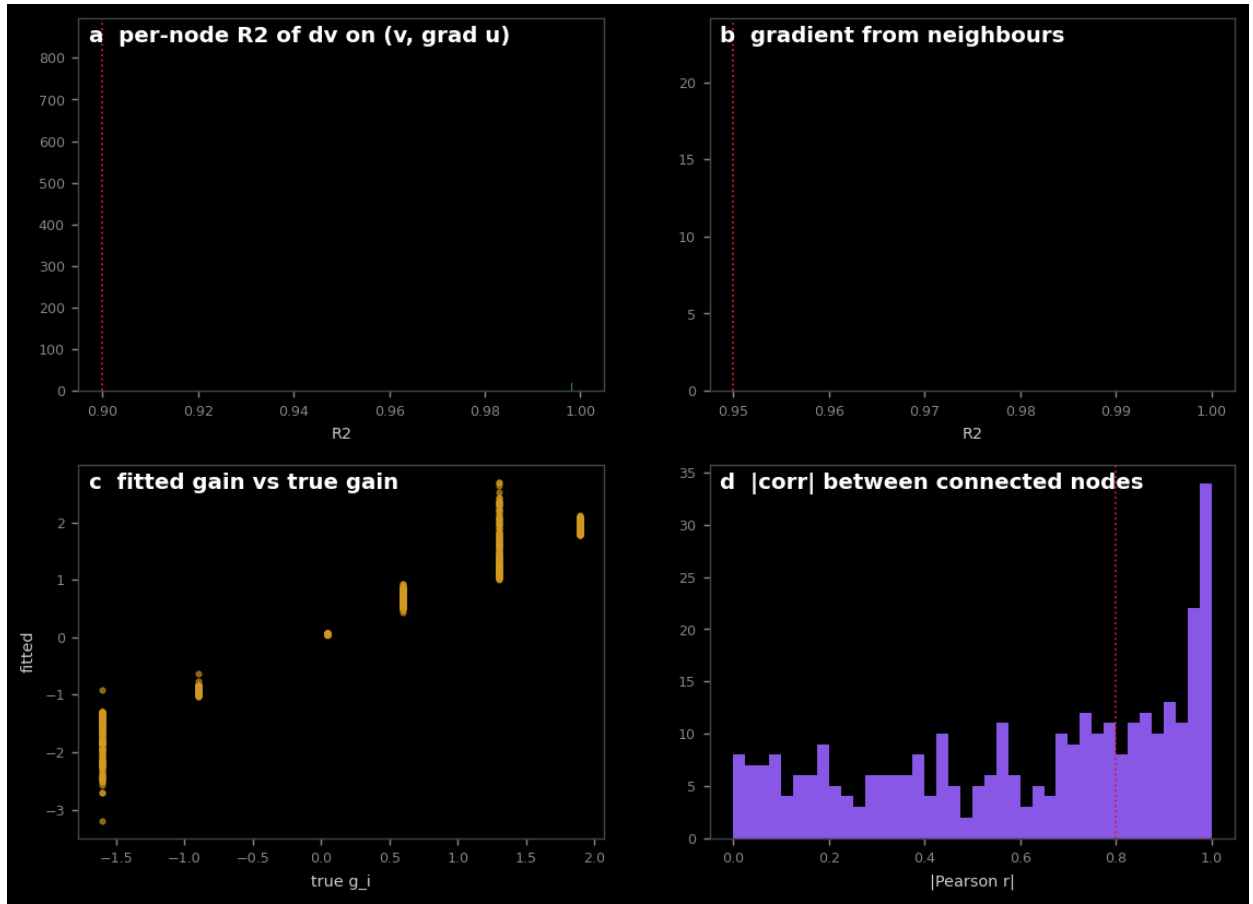


Figure 8: **G25** — connected nodes are not collinear. Threshold: mean $|\text{corr}|$ between connected nodes < 0.80 . Measured: 0.6074. Outcome: PASS.